

Name:	
Class:	



TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION 2017 MATHEMATICS

General Instructions:

- Reading Time: 5 minutes.
- Working Time: 3 hours.
- Write in black pen.
- Board approved calculators & templates may be used
- A Standard Reference Sheet is provided.
- In Question 11 - 16, show all relevant mathematical reasoning and/or calculations.
- Marks may not be awarded for careless or badly arranged working.

Total Marks 100

Section I: 10 marks

- Attempt Question 1 – 10.
- Answer on the Multiple Choice answer sheet provided.
- Allow about 15 minutes for this section.

Section II: 90 Marks

- Attempt Question 11 - 16
- Answer on lined paper provided. Start a new page for each new question.
- Allow about 2 hours & 45 minutes for this section.

The answers to all questions are to be returned in separate *stapled* bundles clearly labelled Question 11, Question 12, etc. Each question must show your Candidate Number.

Section I Multiple Choice (10 marks)

Attempt Question 1 – 10 (1 mark each)

Allow approximately 15 minutes for this section.

1. What is the size of the angle subtended at the centre of a circle with diameter 7cm by an arc 3cm long?

- A. 25° B. 49° C. 67° D. 133°

2. What is the limiting sum of the series $45 - 15 + 5 - \frac{5}{3} + \dots$

- A. $\frac{135}{2}$ B. $\frac{45}{4}$ C. $\frac{135}{4}$ D. $\frac{-45}{2}$

3. Evaluate $\int_{-3}^2 3x^2 - 4x + 1 \, dx$:

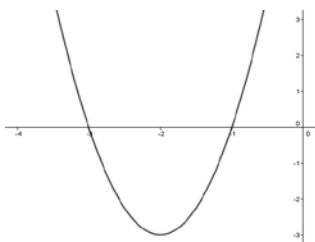
- A. -10 B. 50 C. 14 D. 46

4. $\frac{3+\sqrt{2}}{3-\sqrt{2}} - 5$ is equivalent to:

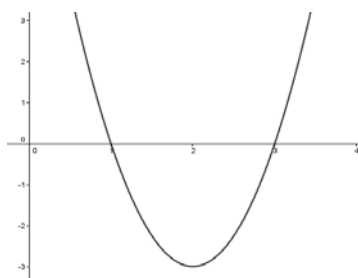
- A. $\frac{-21(2+\sqrt{2})}{7}$ B. $\frac{2(23+3\sqrt{2})}{7}$ C. $\frac{6(5\sqrt{2}-8)}{7}$ D. $\frac{6(\sqrt{2}-4)}{7}$

5. The graph of the gradient function of $y = x^3 + 6x^2 + 9x$ is:

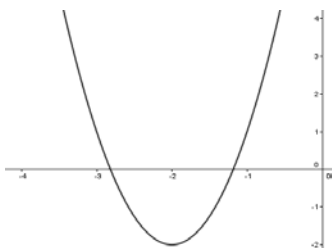
A.



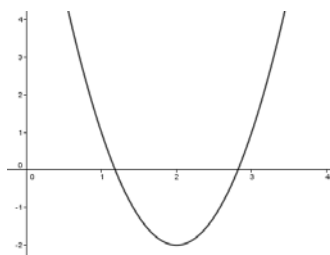
B.



C.



D.



6. John has 10 marbles in a bag, 3 of which are green and the rest are yellow. He randomly draws 1 out of the bag, notes its colour and then puts it back. What is the least number of times John will have to draw so that he has over 95% chance of getting a green marble at least once?

A. 8 B. 9 C. 2 D. 3

7. Find the locus of a point that is equidistant from the point (0,3) and the line $x = -1$

A. $(y - 3)^2 = 2(x + \frac{1}{2})$ B. $x^2 = 8(y + 1)$
C. $x^2 = -8(y + 1)$ D. $(y - 3)^2 = -2(x + \frac{1}{2})$

8. A hundred students in a particular cohort are required to choose at least 1 of 2 languages as a subject. There are 55 students who chose Japanese as a subject and 83 students chose Italian as a subject. A student is randomly selected out of the 100, what is the probability of the student doing both languages?

A. $\frac{28}{100}$ B. $\frac{28}{138}$
C. $\frac{38}{100}$ D. $\frac{38}{138}$

9. The solution to $|3x - 1| = 2x - 1$

A. $x = 0$ and $x = \frac{2}{5}$ B. $x = 0$ only C. $x = \frac{2}{5}$ only D. No Solutions

10. The displacement x metres of a particle moving in a straight line at time t seconds is given by:

$$x = 2t - 4 \log_e(2t + 1).$$

Which one of the following statements is false?

A. The acceleration is always positive B. The initial velocity is -6ms^{-1}
C. The particle is initially at the origin D. As $t \rightarrow \infty, x \rightarrow \infty$ and $v \rightarrow \infty$

End of Section I

Section II Total Marks is 90

Attempt Questions 11 – 16.

Allow approximately 2 hours & 45 minutes for this section.

Answer all questions, starting each new question on a new sheet of paper with your **student ID number** in the top right hand corner and the question number on the left hand side of your paper.

All necessary working must be shown in each and every question.

Question 11 (15 marks)

a) Find the derivative of:

i. $(x + 3)e^{2x}$ (Factorise your answer) 2

ii. $\frac{1+e^x}{\ln(x)}$ 2

b) Let α and β be the roots of the quadratic equation $3x^2 - 7x + 9 = 0$.

i. Find the value of $\alpha^2 + \beta^2$. 2

ii. Factorise $\alpha^3 + \beta^3$ 1

iii. Hence or otherwise, find the value of $\alpha^3 + \beta^3$ 1

c) A locus is given by the equation $x^2 + 2y = 8x - y^2 - 8$.

i. By first completing the square, describe the above locus geometrically. 3

ii. Find the x and y intercepts (if any) of the locus as exact values. 2

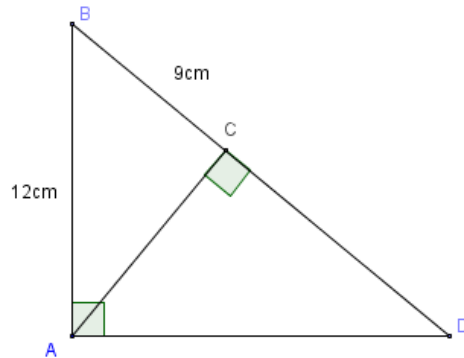
iii. Neatly sketch the locus on a number plane, showing all necessary information. Clearly mark all of the intercepts. 2

Question 12 (15 marks) Start a new sheet of paper

- a) A, B and C are points on the number plane with coordinates (0, 6), (3, 0) and (9, 3) respectively.
- i. Plot the information above on a number plane. (You may add further information on the diagram as the question progresses) 1
 - ii. Find the coordinates of M, the midpoint of AC. 1
 - iii. Show that the equation of AC is given by $x + 3y - 18 = 0$ and the equation of l_1 , the perpendicular bisector of AC, is given by $3x - y - 9 = 0$. 4
 - iv. Find the coordinates of D above AC, such that it lies on l_1 and is $\sqrt{10}$ units from AC. 3
 - v. Show that B, M and D are collinear. 1
 - vi. Without finding the actual distance between A,B, C and D, explain why ABCD is a kite. 1
 - vii. Find the area of ABCD. 3
 - viii. Find the acute angle l_1 makes with the x -axis to the nearest degree. 1

Question 13 (15 marks) Start a new sheet of paper

- a) In the diagram, $\triangle ADC$ and $\triangle ABD$ are both right angle triangles, with $BC = 9\text{cm}$ and $AB = 12\text{cm}$.



- i. Prove that $\triangle ABD \sim \triangle CAD$ 2
 - ii. Hence find the length of DC and AD. 3
- b) Solve $\log_2 x = 3 - \log_2(x - 2)$ 3
- c) Solve $3\sin^2\theta = \sin\theta$ for $0 \leq \theta \leq 2\pi$ (Give your answer in radians) 3
- d)
- i. Sketch a graph of the region bounded by the curves $y = x^2$ and $y = x^3$, clearly show their points of intersection at $(0, 0)$ and $(1, 1)$. 2
 - ii. Find the volume of the solid that is formed when the region in part i) rotated about the x -axis. 2

Question 14 (15 marks) Start a new sheet of paper

- a) The population P of a small town is decreasing according to the equation $\frac{dP}{dt} = -kP$, where time t is the number of years since the beginning of May 2015 and k is a positive constant.

At the start of May 2015 the town had a population of 2040. However, the population had halved by the beginning of May 2017.

It is given that $P = P_0 e^{-kt}$ is a solution of $\frac{dP}{dt} = -kP$.

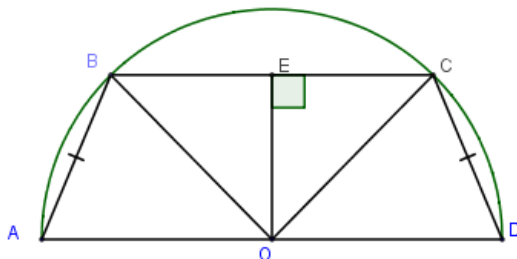
- i. Find the value of P_0 . 1
 - ii. Show that the value of k is $\frac{1}{2} \ln 2$. 2
 - iii. The town mayor decrees that when there are only 100 people left then they must all leave. During which month of which year will this happen? 3
- b) A particle initially at $x = -2$, moves so that at time t seconds its velocity v m/s is given by:

$$v = 2\pi + 2\pi(\sin \pi t).$$

- i. Find the initial velocity. 1
 - ii. Find the time when the particle is first stationary, and its displacement at that time. 3
 - iii. Sketch a velocity-time graph for the first 2 seconds. 2
- c) Two employees are paid daily in two different schemes. The 1st employee receives a payment that is 3 times more than what he was paid the day before. The 2nd employee receives a payment that is \$50 dollars more than what he was paid the day before. Both employees were paid the same amount of money on day 1 and the same amount in total after 8 days of work. How much did they get paid on day 1? Write your answer to the nearest cent. 3

Question 15 (15 marks) Start a new sheet of paper

- a) An isosceles trapezium ABCD is drawn with its vertices on a semi-circle centre O and diameter 20 cm (see diagram). OE is the altitude of ABCD.



- i. Prove that $\triangle BOE \equiv \triangle COE$ 2
 - ii. Hence or otherwise, show that the area of the trapezium ABCD is given by: 3

$$A = \frac{1}{4}(x + 20)\sqrt{400 - x^2}$$

where x is the length of BC.
 - iii. Hence find the length of BC so that the area of the trapezium ABCD is a maximum. 3
- b) An unbiased coin is tossed 3 times, and the upper most face is recorded after each toss.
- i. Draw a tree diagram to illustrate all the possible outcomes 1
 - ii. What is the difference in probability between getting exactly two heads and getting the same outcome for all three tosses? 3
- c) Use the trapezoidal rule with 2 sub-intervals to estimate the value of $\int_{-2}^2 x e^x dx$ correct to 1 decimal place. 3

PLEASE TURN OVER

Question 16 (15 marks) Start a new sheet of paper

- a) Hercules walks at a speed of 3km/h at a bearing of 030T. Two hours later from the same spot, Xena walks at a speed of 5km/h at a bearing of 160T.
- i. Write an expression for the distance Hercules is from the origin t hours after Xena started her walk. 1
 - ii. Find the distance between Hercules and Xena 8 hours after Xena started her walk. 3
- b) A home loan of \$600 000 is borrowed from a bank at a rate of 6% *p. a.* on the 1st of January 2017 over 10 years, compounded monthly. Interest is charged at the beginning of each month (including immediately when the loan was taken), while repayments are made at the end of every month. Let M be the amount of each monthly instalment.
- Let I_n represent the amount owing after the n^{th} repayment.
- i. Show that $I_n = 600000\left(\frac{201}{200}\right)^n - 200M\left[\left(\frac{201}{200}\right)^n - 1\right]$ 3
 - ii. The loan is to be paid back over 10 years. Show that $M = 6661.23$ 2
 - iii. The bank reduced its interest rates from 6% *p. a.* to 5% *p. a.* on the 1st of January 2022. Find, to the nearest cent, the amount owing just before the rate changes. 1
 - iv. Denote your answer in part iii. as ' A ' for this question.

If the repayment stays the same and taking into account the rate change, in which year and month can the loan be expected to be paid off? 3

Hint: Do all your working out in terms of A and M , only substitute your answers from ii. and iii. at the end.
 - v. Due to the rate change in part iii, the bank requires the last payment to only pay off the remaining balance as opposed to the usual instalment amount. How much interest in total will the borrower have paid over the life of the loan? 2

End of Paper

Student Number MC

1. $l = 10$

$$\theta = \frac{l}{r} = \frac{3}{3.5} = \frac{6}{7}$$

$$\theta \text{ in degrees} = 49^\circ$$

(B)

2. $a = 45$ $r = \frac{1}{3}$

$$S_\infty = \frac{45}{1 - (\frac{1}{3})} = \frac{45}{\frac{2}{3}} = \frac{135}{2}$$

(C)

3.
$$\left[x^2 - 2x^2 + x \right]_{-3}^2 = (8 - 8 + 2) - (-27 - 18 - 3)$$
$$= 2 - (-48)$$
$$= 50$$

(B)

4.
$$\frac{3 + \sqrt{2}}{3 - \sqrt{2}} - 5 = \frac{3 + \sqrt{2} - 15 + 5\sqrt{2}}{3 - \sqrt{2}}$$

$$= \frac{-12 + 6\sqrt{2}}{3 - \sqrt{2}}$$

$$= \frac{(-12 + 6\sqrt{2})(3 + \sqrt{2})}{9 - 2}$$

$$= \frac{-36 - 12\sqrt{2} + 18\sqrt{2} + 12}{7}$$

$$= \frac{-24 + 6\sqrt{2}}{7}$$

$$= \frac{6(\sqrt{2} - 4)}{7}$$

(D)



Student Number _____

5. $\frac{dy}{dx} = 3x^2 + 12x + 9$

$$= 3(x^2 + 4x + 3)$$

$$= 3(x+3)(x+1)$$

(A)

6. $P(\text{green}) = \frac{3}{10}$ $P(\text{yellow}) = \frac{7}{10}$

$$P(\text{at least 1 green}) = 1 - P(\text{All yellow})$$
$$= 1 - \left(\frac{7}{10}\right)^n$$

$$\therefore \text{Solving for } 1 - \left(\frac{7}{10}\right)^n > \frac{19}{20}$$
$$\left(\frac{7}{10}\right)^n < \frac{1}{20}$$

$$\ln \left(\frac{7}{10}\right)^n < \ln \left(\frac{1}{20}\right)$$

$$n \ln \left(\frac{7}{10}\right) < \ln \left(\frac{1}{20}\right)$$

$$n > \frac{\ln \left(\frac{1}{20}\right)}{\ln \left(\frac{7}{10}\right)}$$

$$\ln \left(\frac{7}{10}\right)$$

$$n > 8.39$$

$$\therefore n = 9$$

(B)

7. $\sqrt{(x-0)^2 + (y-3)^2} = (x+1)$

$$x^2 + (y-3)^2 = x^2 + 2x + 1$$

$$(y-3)^2 = 2x + 1$$

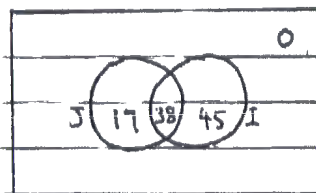
$$(y-3)^2 = 2\left(x + \frac{1}{2}\right)$$

(A)



Student Number

8.



$$\therefore P(\text{Both}) = \frac{38}{100}$$

9. Solve $(3x-1)^2 = (2x-1)^2$
 $9x^2 - 6x + 1 = 4x^2 - 4x + 1$

$$5x^2 - 2x = 0$$

$$x(5x - 2) = 0$$

$$\therefore x = 0 \text{ or } \frac{2}{5}$$

check $x=0$, LHS = 1, RHS = -1, Incorrect.

$x = \frac{2}{5}$, LHS = $\left|\frac{6}{5} - 1\right| = \frac{1}{5}$, RHS = $\frac{4}{5} - 1 = -\frac{1}{5}$. $\therefore$ Incorrect

$\therefore$ No solutions

(D)

10. $x = 2t - 4\ln(2t+1)$ $x(0) = 0$

$$v = 2 - \frac{4}{2t+1} \times 2$$

$$v(0) = -6$$

$$a > 0$$

$$v = 2 - \frac{8}{4t+1}$$

as $t \rightarrow \infty$, $v \rightarrow 2$, $a \rightarrow 0$.

$$a = 8(4t+1)^{-2} \cdot 4$$

$$= 32$$

$$(4t+1)^2$$

$\therefore$ (D)

MATHEMATICS: Question...11...

Suggested Solutions

Marks Awarded

Marker's Comments

a) Derivative of

(i) $(x+3)e^{2x}$

$$\frac{dy}{dx} = (x+3)2e^{2x} + e^{2x} \quad \checkmark$$

$$= e^{2x}(2x+6+1)$$

$$= e^{2x}(2x+7) \quad \checkmark$$

2 marks

✓ process

✓ answer.

let $u = (x+3)$

$u' = 1$

$v = e^{2x}$

$v' = 2e^{2x}$

(ii) $\frac{1+e^x}{\ln x}$

$$\frac{dy}{dx} = \frac{vu' - uv'}{v^2}$$

$$= \frac{\ln x e^x - (1+e^x)\frac{1}{x}}{(\ln x)^2}$$

$$= \frac{x e^x \ln x - 1 - e^x}{x (\ln x)^2}$$

2 marks

✓ sub
in correct
formula

✓ Simplifying
Had to
eliminate
fractions
from
numerator
to simplify

$u = 1+e^x$

$u' = e^x$

$v = \ln x$

$v' = \frac{1}{x}$

- accepted

$$\frac{e^x(x \ln x - 1) - 1}{x (\ln x)^2}$$

b(i) $3x^2 - 7x + 9 = 0$

$$\alpha + \beta = \frac{-b}{a} = \frac{7}{3}$$

$$\alpha \beta = \frac{c}{a} = 3$$

$$(i) \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= \left(\frac{7}{3}\right)^2 - 2(3)$$

$$= \frac{49}{9} - 6$$

$$= -\frac{5}{9}$$

2 marks

✓ Simplifying

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

✓ getting

$-\frac{5}{9}$ as

answer

well done

Suggested Solutions

Marks Awarded

Marker's Comments

b(ii) Factorise $\alpha^3 + \beta^3$

$$\alpha^3 + \beta^3 = (\alpha + \beta)(\alpha^2 - \alpha\beta + \beta^2) \quad \checkmark$$

1 mark.

(iii) Value $\alpha^3 + \beta^3$

$$\begin{aligned} \alpha^3 + \beta^3 &= \left(\frac{7}{3}\right) \times \left(-\frac{5}{9} - 3\right) \\ &= \frac{7}{3} \times \frac{-32}{9} \\ &= -\frac{224}{27} \quad \checkmark \end{aligned}$$

1 mark

c) $x^2 + 2y = 8x - y^2 - 8$

$$\begin{aligned} (1) \quad x^2 - 8x + y^2 + 2y + 8 &= 0 \\ x^2 - 8x + 16 + y^2 + 2y + 1 &= -8 + 16 + 1 \\ (x-4)^2 + (y+1)^2 &= 9 \end{aligned}$$

locus is a circle with centre (4, -1) and radius 3

3 marks)

✓ correct completion of square
✓ getting $(x-4)^2 + (y+1)^2 = 9$
✓ writing locus as circle with radius.

(ii) y-intercept put $x=0$

$$\begin{aligned} (x-4)^2 + (y+1)^2 &= 9 \\ 4^2 + (y+1)^2 &= 9 \\ (y+1)^2 &= -7 \end{aligned}$$

∴ No y-intercepts ✓

x-intercept put $y=0$

$$\begin{aligned} (x-4)^2 + (0+1)^2 &= 9 \\ (x-4)^2 + 1 &= 9 \\ (x-4)^2 &= 8 \\ x-4 &= \pm\sqrt{8} \\ x &= 4 \pm \sqrt{8} \text{ or } 4 \pm 2\sqrt{2} \quad \checkmark \end{aligned}$$

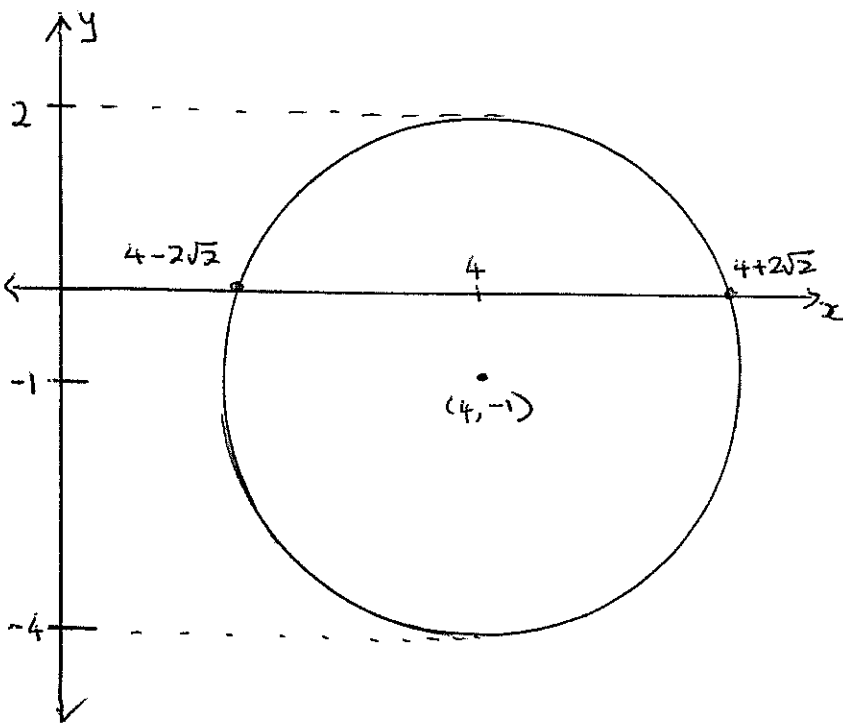
2 marks

✓ there is no y-intercepts
✓ x-intercepts

Suggested Solutions

Marks Awarded

Marker's Comments



2 marks

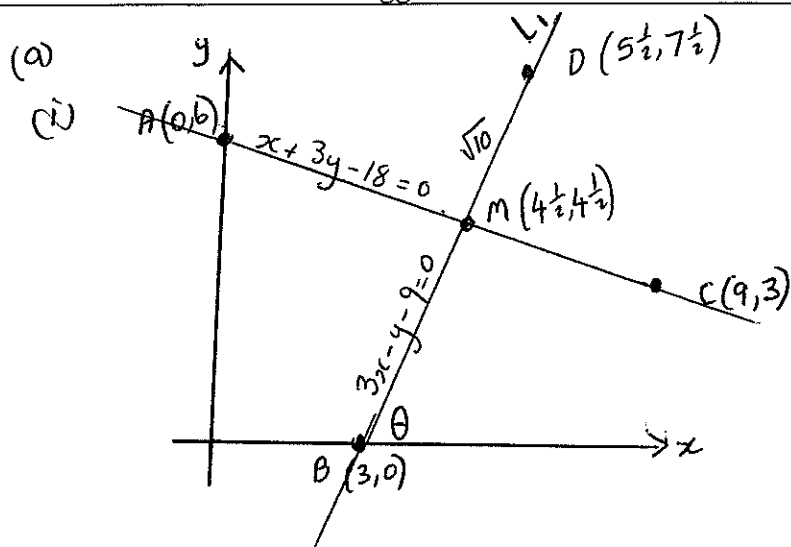
✓ shape
centre point

✓ x-intercept
and maximum/minimum
y values
ie 2 and -4

Suggested Solutions

Marks

Marker's Comments



①

A, B, C labeled
x-y axis labeled
axis roughly to
scale.

NOTE! Add any other
features as you progress
through the question.

(ii) $M = \left(\frac{9+0}{2}, \frac{6+3}{2} \right)$

$M = (4\frac{1}{2}, 4\frac{1}{2})$. (midpoint of AC)

①

(iii) To find equation of AC

$m_{AC} = \frac{3-6}{9-0} = \frac{-3}{9} = -\frac{1}{3}$

Eqn of AC: using $y = mx + b$
 $y = -\frac{1}{3}x + 6$
 $x + 3y - 18 = 0$

①

Alternatively,
 $\frac{y-6}{x-0} = \frac{6-3}{0-9}$

$\frac{y-6}{x} = -\frac{1}{3}$

$3y - 18 = -x$

$x + 3y - 18 = 0$

①

Note! You needed to show
all steps clearly as the
answer was given. or you
do not get the
mark.

Since L_1 is perpendicular then
 $m_1 m_2 = -1$. L_1 passes through $M(\frac{9}{2}, \frac{9}{2})$

$\therefore m_{L_1} = 3$

$\therefore$ Eqn of L_1 : $y - \frac{9}{2} = 3(x - \frac{9}{2})$
 $2y - 9 = 6(x - \frac{9}{2})$
 $2y - 9 = 6x - 27$
 $6x - 2y - 18 = 0$

①

①

must show working
because once again
the answer is
given.

(IV) Let the coordinates of D be (x, y)

The perpendicular distance of D from

AC ($x+3y-18=0$) is $\sqrt{10}$.

$$\therefore \sqrt{10} = \frac{|x+3y-18|}{\sqrt{1+9}}$$

$$10 = |x+3y-18|$$

$$x+3y-18=10 \text{ OR } x+3y-18=-10$$

$$x+3y-28=0$$

$$y = -\frac{x}{3} + \frac{28}{3}$$

$$x+3y-8=0$$

$$y = -\frac{x}{3} + \frac{8}{3}$$

Since D is above the line then

$x+3y-28=0$ is the line

$\sqrt{10}$ units from AC. To find the

point D, solve

$x+3y-28=0$ ① and ② simultaneously.

$$3x-y-9=0 \text{ --- ②}$$

$$\textcircled{1} \times 3 \quad 3x+9y-84=0 \text{ --- ①} \times 3$$

$$- \textcircled{2} \quad - (3x-y-9=0) \text{ --- ②}$$

$$10y-75=0$$

$$y = \frac{75}{10}$$

$$y = \frac{15}{2} \text{ or } 7\frac{1}{2}$$

When $y = \frac{15}{2}$ substitute into ①

$$x+3\left(\frac{15}{2}\right)-28=0$$

$$x+\frac{45}{2}-28=0$$

$$x = \frac{11}{2} \text{ or } 5\frac{1}{2}$$

$$\therefore D = \left(5\frac{1}{2}, 7\frac{1}{2}\right)$$

①

For using perpendicular distance of point from a line formula & substituting correctly.

①

①

Suggested Solutions

Marks

Marker's Comments

(v) To show B, M & D are collinear
 substitute B into eqn l_1 . $B = (3, 0)$
 $3(3) - (0) - 9 = 0$
 $\therefore B(3, 0)$ lies on the line l_1 .
 $\therefore B, M$ and D are collinear.

①

Alternatively

$$m_{BM} = \frac{4\frac{1}{2} - 0}{4\frac{1}{2} - 3} = \frac{4\frac{1}{2}}{1\frac{1}{2}} = 3$$

$$m_{MD} = \frac{7\frac{1}{2} - 4\frac{1}{2}}{5\frac{1}{2} - 4\frac{1}{2}} = \frac{3}{1} = 3.$$

Since $m_{BM} = m_{MD}$ then
 B, M and D are collinear.

Alternate ①

(vi) Since l_1 is the perpendicular bisector of AC , then $ABCD$ is a kite. (One diagonal is the perpendicular bisector of the other).

①

Note! It is not sufficient to say the diagonals are perpendicular. Must also mention bisector.

(vii)

$$\begin{aligned} d_{AC} &= \sqrt{(9-0)^2 + (3-6)^2} \\ &= \sqrt{81+9} \\ &= \sqrt{90} \end{aligned}$$

①

$$\begin{aligned} d_{BD} &= \sqrt{\left(3-\frac{11}{2}\right)^2 + \left(0-\frac{15}{2}\right)^2} \\ &= \sqrt{\frac{25}{4} + \frac{225}{4}} \\ &= \sqrt{\frac{250}{4}} \\ &= \frac{5\sqrt{10}}{2} \end{aligned}$$

①

$$\begin{aligned} \text{Area } ABCD &= \frac{1}{2} \times AC \times BD \\ &= \frac{1}{2} \times \sqrt{90} \times \frac{5\sqrt{10}}{2} \\ &= \frac{1}{2} \times 3\sqrt{10} \times \frac{5\sqrt{10}}{2} \\ &= \frac{150}{4} u^2 \quad \text{OR} \quad 37\frac{1}{2} u^2 \end{aligned}$$

①

$$\tan \theta = \frac{7\frac{1}{2}}{2\frac{1}{2}} \quad \left(\text{or } \frac{4\frac{1}{2}}{1\frac{1}{2}}\right)$$

(viii)

$$\tan \theta = 3$$

$$\theta = \tan^{-1} 3$$

$$= 72^\circ \quad (\text{nearest degree})$$

①

Alternatively you could find the areas using triangles.

$$\text{eg } AC = 3\sqrt{10}$$

$$BM = \frac{15}{\sqrt{10}}$$

$$\text{Area} = \text{Area } \triangle ADC + \text{Area } \triangle ABC$$

$$\begin{aligned} &= \frac{1}{2} \times 3\sqrt{10} \times \sqrt{10} \\ &\quad + \frac{1}{2} \times 3\sqrt{10} \times \frac{15}{\sqrt{10}} \end{aligned}$$

$$\begin{aligned} &= 15 + \frac{45}{2} \\ &= 37\frac{1}{2} \end{aligned}$$

Note! Answer in radians, no marks. (make sure your calculator is in correct mode).

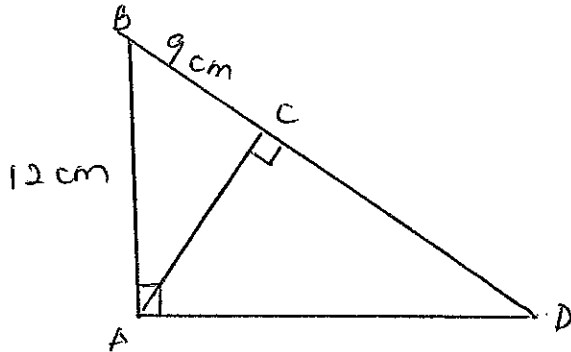
71° accepted if $\tan \theta = 3$ is shown.

Suggested Solutions

Marks Awarded

Marker's Comments

ai) In $\triangle ABD$ and $\triangle CAD$



$\angle BAD = \angle ACD = 90^\circ$ (given) } — 1
 $\angle D$ is common.

$\therefore \triangle ABD \sim \triangle CAD$ (equiangular) — 1

ii) $9^2 + AC^2 = 12^2$ (Pythagoras' Theorem)

$$AC^2 = 144 - 81$$

$$= \sqrt{63} \quad (AC > 0) \quad *$$

$$= 3\sqrt{7}$$

$$\frac{\triangle ABD}{\triangle CAD} = \frac{AB}{CA} = \frac{BD}{AD} = \frac{AD}{CD}$$

corresponding sides in similar triangles are in same ratio

$$\frac{12}{\sqrt{63}} = \frac{9+CD}{AD} = \frac{AD}{CD}$$

$$AD = \frac{12CD}{\sqrt{63}} \quad \text{--- i}$$

$$AD^2 = CD(9+CD) \quad \text{--- ii}$$

sub i into ii

$$\left(\frac{12CD}{\sqrt{63}}\right)^2 = 9CD + CD^2 \quad \text{--- iii}$$

$$\frac{144CD^2}{63} = 9CD + CD^2$$

please state

some students wrote are equal

1 — correct reasons

2. — to find AC, AD, CD

Suggested Solutions

Marks Awarded

Marker's Comments

$$144 CD^2 = 567CD + 63CD^2$$

$$81CD^2 = 567CD$$

$$CD = 0 \text{ or } 7 \quad (CD > 0)$$

$$\therefore CD = 7$$

sub into eqn i

$$AD = \frac{12(7)}{\sqrt{63}}$$

$$= \frac{84}{\sqrt{63}} \text{ or } \frac{28}{\sqrt{7}} \text{ or } 4\sqrt{7}$$

$$\text{or } 10.58 \text{ or } \sqrt{112}.$$

OTHER METHODS.

1) trig

$$\cos B = \frac{9}{12}$$

$$\therefore B = 41.41^\circ$$

$$\angle D = 90 - 41.41$$

$$= 48.59^\circ$$

$$\sin 48.59 = \frac{AC}{AD}$$

$$AD = \frac{\sqrt{63}}{\sin 48.59}$$

$$= 10.58.$$

$$AD^2 + AB^2 = BD^2 \quad (\text{pythagoras theorem})$$

$$10.58^2 + 12^2 = BD^2$$

$$BD = 15.998$$

$$CD = BD - BC$$

$$= 15.998 - 9 = 7 \text{ or } 6.998$$

* some used
sine rule

* some used
pythagoras
all the way
through.

Suggested Solutions

Marks Awarded

Marker's Comments

$$b) \log_2 x = 3 - \log_2 (x-2)$$

$$\log_2 x = \log_2 2^3 - \log_2 (x-2) \quad \text{---} \quad 1$$

$$\log_2 x = \log_2 \left(\frac{8}{x-2} \right)$$

$$x = \frac{8}{x-2}$$

$$x^2 - 2x - 8 = 0$$

$$\begin{array}{l} x-4 \\ x+2 \end{array} \quad (x-4)(x+2) = 0$$

$$x = -2 \text{ or } 4 \quad \text{---} \quad 1$$

$$\text{but } x > 0 \therefore x = 4 \text{ only} \quad \text{---} \quad 1$$

$$c) 3\sin^2 \theta = \sin \theta \text{ for } 0 \leq \theta \leq 2\pi$$

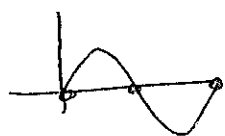
$$3\sin^2 \theta - \sin \theta = 0$$

$$\sin \theta (3\sin \theta - 1) = 0 \quad \text{---} \quad 1$$

$$\sin \theta = 0 \quad 3\sin \theta = 1$$

$$\therefore \theta = 0, \pi, 2\pi \quad \sin \theta = \frac{1}{3} \quad \begin{array}{c} \checkmark \\ + \\ \checkmark \end{array} \quad \text{---} \quad 1$$

$$\theta = 0.34, 2.8 \quad \text{---} \quad 1$$



$$\therefore \theta = 0, 0.34, 2.8, \pi, 2\pi$$

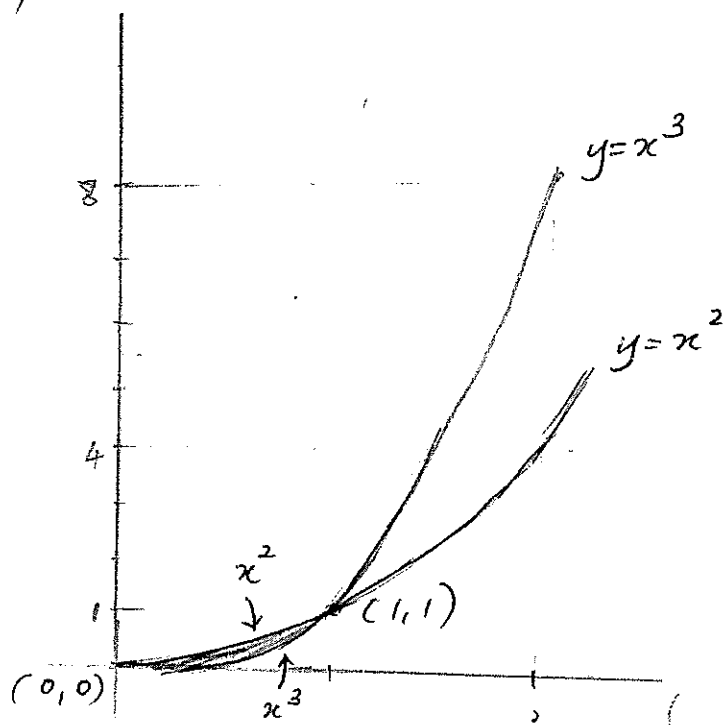
* DO NOT
divide the eqn
by $\sin \theta$.

Suggested Solutions

Marks Awarded

Marker's Comments

d) i)



1 — shape
1 — positioning correctly.

* — some students had the graph of x^3 higher than x^2 in the bounded region.
— need not sketch the left side
* must show intersection pts.

$$\begin{aligned} \text{ii)} \quad V &= \pi \int_0^1 (x^2)^2 - (x^3)^2 \\ &= \pi \left[\int_0^1 x^4 - \int_0^1 x^6 \right] \\ &= \pi \left(\left[\frac{x^5}{5} \right]_0^1 - \left[\frac{x^7}{7} \right]_0^1 \right) \\ &= \frac{2\pi}{35} \end{aligned}$$

mistakes!
 $(x^2 - x^3)^2$
 $(x^3)^2 = x^6$

Suggested Solutions

Marks Awarded

Marker's Comments

$$a) i) P = P_0 e^{-kt} \quad \frac{dP}{dt} = -kP_0 e^{-kt}$$

$$= -kP$$

$$\text{when } t = 0, P = 2040.$$

$$\therefore 2040 = P_0 e^{-0} :$$

$$\therefore P_0 = 2040.$$

$$ii) t = 2 \quad P = 1020.$$

$$1020 = 2040 e^{-2k}.$$

$$e^{-2k} = \frac{1}{2}$$

$$e^{2k} = 2.$$

$$2k = \ln 2$$

$$k = \frac{1}{2} \ln 2$$

$$iii) P = 100$$

$$100 = 2040 e^{-\frac{1}{2} \ln 2 (t)}$$

$$\frac{5}{102} = e^{-\frac{t}{2} \ln 2}$$

$$\frac{t}{2} \ln 2 = \ln \frac{5}{102}$$

$$t = 8.7 \text{ years.}$$

show some working please!

Suggested Solutions

Marks Awarded

Marker's Comments

$$= 8 \text{ years and } 0.7 \times 12 \\ 8.4 \text{ months}$$

$$2015 + 8 = 2023$$

$$\text{May} + 8.4 \text{ mths} = \text{Jan } 0.4 \times 31 \approx 12.4$$

$\therefore$ During January 2024 — 1

b) i) when $t = 0$ for initial condition

$$\therefore v = 2\pi + 2\pi(\sin \pi(0))$$

$$= 2\pi \text{ m/s}$$

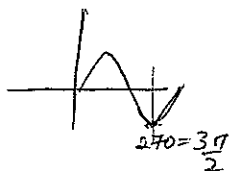
ii) first stationary $v = 0$.

$$2\pi + 2\pi \sin \pi t = 0$$

$$\sin \pi t = -1$$

$$\pi t = \frac{3\pi}{2}$$

$$t = \frac{3}{2}$$



$$x = \int 2\pi + 2\pi \sin \pi t$$

$$= 2\pi t - 2 \cos \pi t + C$$

when $t = 0$, $x = -2$

$$-2 = 0 - 2(1) + C$$

$$\therefore C = 0$$

$$\therefore x = 2\pi t - 2 \cos \pi t$$

when $t = \frac{3}{2}$

$$x = 2\pi \left(\frac{3}{2}\right) - 2 \cos \pi \left(\frac{3}{2}\right)$$

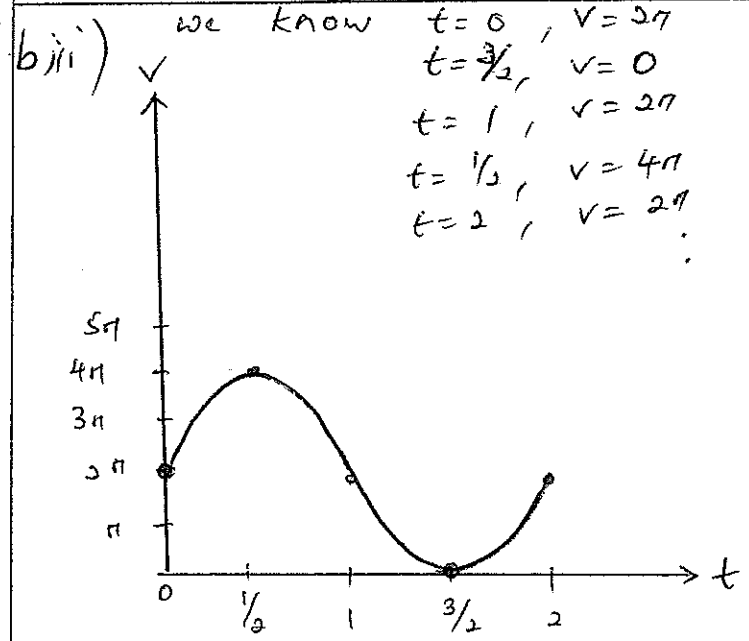
$$= 3\pi - 0$$

$$= 3\pi \text{ metres}$$

Suggested Solutions

Marks Awarded

Marker's Comments



c. Employee 1
 $a = x, T_2 = 3x, T_3 = 9x$
 $T_n = ar^{n-1}$
 $T_8 = x3^7$
 $= 2187x$
 $S_n = \frac{a(r^n - 1)}{r - 1}$
 $S_8 = \frac{x(3^8 - 1)}{3 - 1}$
 $= 3280x$

Employee 2
 $T_1 = x, T_2 = x + 50$
 $\therefore d = 50, n = 8$
 $T_n = a + (n - 1)d$
 $T_8 = x + (7)50$
 $= x + 350$
 $S_n = \frac{n}{2}(a + l)$
 $S_8 = \frac{8}{2}(x + (x + 350))$
 $= 4(2x + 350)$
 $= 8x + 1400$

Equate both
 $3280x = 8x + 1400$
 $3272x = 1400$
 $x = 0.43 \text{ (2 d.p.)}$

1

1

1

Q(i) Prove $\triangle BOE \equiv \triangle COE$

In $\triangle BOE$ and $\triangle COE$

$$\angle CEO = \angle BEO = 90^\circ \text{ (given)}$$

EO is common

BO = CO (radii of a circle)

$\therefore \triangle BOE \equiv \triangle COE$ (RHS)

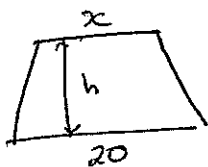
2 marks

Accepted SSS

- Circle property
perpendicular line
from centre of
circle bisects
the chord.

* For SAS
only accepted
if mentioned
the above circle
property

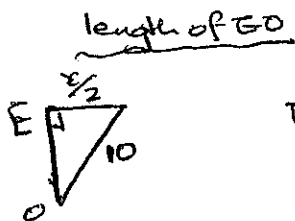
(ii)



Let BC be x

$$\text{Area of trapezium} = \frac{1}{2}(a+b)h$$

$$= \frac{1}{2}(x+20) \times EO \quad \checkmark$$



$$EO^2 + \left(\frac{x}{2}\right)^2 = 10^2$$

$$EO^2 = 100 - \frac{x^2}{4}$$

$$EO^2 = \frac{400 - x^2}{4}$$

$$EO = \frac{\sqrt{400 - x^2}}{2} \quad (EO > 0) \quad \checkmark$$

$$\therefore A = \frac{1}{2}(x+20) \times \frac{\sqrt{400 - x^2}}{2}$$

$$= \frac{1}{4}(x+20)\sqrt{400 - x^2} \quad \checkmark$$

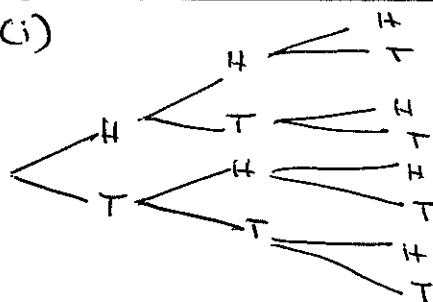
(iii)	Suggested Solutions	Marks Awarded	Marker's Comments								
	length of BC so that area is maximum $\therefore \frac{dA}{dx} = \frac{1}{4}(x+20) \cdot \frac{1}{2}(400-x^2)^{-\frac{1}{2}} \cdot (-2x) + \sqrt{400-x^2}$ $= \frac{1}{4} \left[\frac{-x(x+20)}{\sqrt{400-x^2}} + \sqrt{400-x^2} \right]$ $= \frac{1}{4} \left[\frac{-x^2-20x+400-x^2}{\sqrt{400-x^2}} \right]$ $= \frac{1}{4} \left[\frac{-2x^2-20x+400}{\sqrt{400-x^2}} \right]$ $= \frac{1}{2} \left[\frac{-x^2-10x+200}{\sqrt{400-x^2}} \right] \checkmark$ $\frac{dA}{dx} = 0$ $-x^2-10x+200=0$ $x^2+10x-200=0$ $(x+20)(x-10)=0$ $x = -20 \text{ or } 10$ $\therefore x=10$ only as $x > 0$ ✓ Test <table><tr><td>x</td><td>9</td><td>10</td><td>11</td></tr><tr><td>$\frac{dA}{dx}$</td><td>0.91</td><td>0</td><td>-0.93</td></tr></table> $\therefore$ maximum at $x=10$	x	9	10	11	$\frac{dA}{dx}$	0.91	0	-0.93		$u = x+20$ so $u' = 1$ $v = \sqrt{400-x^2}$ $\therefore v' = \frac{1}{2}(400-x^2)^{-\frac{1}{2}} \cdot (-2x)$ $= \frac{-x}{\sqrt{400-x^2}}$
x	9	10	11								
$\frac{dA}{dx}$	0.91	0	-0.93								

Suggested Solutions

Marks Awarded

Marker's Comments

b(i)



(ii) $P(2 \text{ heads}) = \frac{3}{8}$ ✓

$P(\text{same outcomes}) = \frac{2}{8} + \frac{1}{4}$ ✓

Difference = $\frac{1}{8}$ ✓

(c) Trapezoidal Rule

-2	0	2
$-2e^{-2}$	0	$2e^2$

$A = \frac{2-0}{2} [-2e^{-2} + 2(0) + 2e^2]$

$= 1 (2e^{-2} + 2e^2)$ ✓

$= -4e^{-2} + 4e^4$

$\approx 14.5 \text{ (to 1 dp)}$ ✓

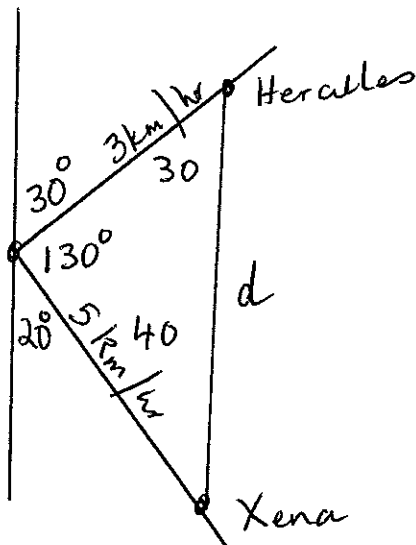
NB 2 sub-intervals only

Suggested Solutions

Marks

Marker's Comments

a)



When Xena starts,
Heraclus has been
walking $(t+2)$ hours.

$$\begin{aligned} (i) \quad D_H &= 3 \times (t+2) \\ &= 3t + 6 \end{aligned}$$

①

(ii) When Xena walks for 8 hours,
Heraclus has been walking for 10 hours.

$\therefore$ When $t=8$

$$D_{\text{Xena}} = 5 \times 8 = 40 \text{ km}$$

①

$$D_{\text{Heraclus}} = 3 \times 10 = 30 \text{ km}$$

Let d be distance between Xena & Heraclus.

$$\begin{aligned} d^2 &= 30^2 + 40^2 - 2 \times 30 \times 40 \times \cos 130^\circ \\ &= 4042.69 \end{aligned}$$

①

$$d = 63.58 \text{ km} \quad d > 0$$

①

Note! if
 $\theta x^2 = 58.15$ then
your calculator is
in radians &
answer is
wrong!!

Suggested Solutions

Marks

Marker's Comments

(b) $P = \$600000$ 6% pa

Compounded monthly $r = \frac{0.06}{12}$

$= \frac{1}{200}$ (or 0.005)

After 1st month

$$I_1 = 600000 \times \left(1 + \frac{1}{200}\right)^1 - M$$

$$= 600000 \times \left(\frac{201}{200}\right)^1 - M$$

$$I_2 = \left[600000 \left(\frac{201}{200}\right)^1 - M\right] \left(\frac{201}{200}\right) - M$$

$$= 600000 \left(\frac{201}{200}\right)^2 - M \left(\frac{201}{200}\right) - M$$

$$I_3 = 600000 \left(\frac{201}{200}\right)^3 - M \left(\frac{201}{200}\right)^2 - M \left(\frac{201}{200}\right) - M$$

⋮

$$I_n = 600000 \left(\frac{201}{200}\right)^n - M \left[\left(\frac{201}{200}\right)^{n-1} + \left(\frac{201}{200}\right)^{n-2} + \dots + \frac{201}{200} + 1 \right]$$

$$= 600000 \left(\frac{201}{200}\right)^n - M \left[1 + \frac{201}{200} + \left(\frac{201}{200}\right)^2 + \dots + \left(\frac{201}{200}\right)^{n-1} \right]$$

↑

This is a GP where $a = 1$ & $r = \frac{201}{200}$

$$\therefore S_n = 1 \left(\frac{\left(\frac{201}{200}\right)^n - 1}{\frac{201}{200} - 1} \right) = 200 \left(\left(\frac{201}{200}\right)^n - 1 \right)$$

$$I_n = 600000 \left(\frac{201}{200}\right)^n - M \left(200 \left(\frac{201}{200}\right)^n - 1 \right)$$

$$= 600000 \left(\frac{201}{200}\right)^n - 200M \left(\left(\frac{201}{200}\right)^n - 1 \right)$$

This is given so you MUST show clear working towards this.

(ii) 10 years = 120 months.

$$\therefore t = 120$$

Loan is paid off when $I_n = 0$

$$\text{ie } 0 = 600000 \left(\frac{201}{200} \right)^{120} - 200M \left(\left(\frac{201}{200} \right)^{120} - 1 \right)$$

①*

$$200M = \frac{600000 \left(\frac{201}{200} \right)^{120}}{\left(\frac{201}{200} \right)^{120} - 1}$$

$$200M = \frac{1091638.04}{0.819396734}$$

$$200M = 1332246.023$$

$$M = \$6661.23$$

$\therefore$ Monthly repayments are \$6661.23.

①*

$$\text{OR } M = \frac{3000 \left(\frac{201}{200} \right)^{120}}{\left(\frac{201}{200} \right)^{120} - 1}$$

(this answer was given, so explanation working needed to be clear to get the marks)

(iii) 1st January 2022 is exactly 5 years (or 60 months).

$$I_{60} = 600000 \left(\frac{201}{200} \right)^{60} - \$6661.23 \times 200 \left[\left(\frac{201}{200} \right)^{60} - 1 \right]$$

$$= 809310.0915 - 464754.2203$$

$$= \$344555.87$$

①*

This is the amount owing before the rate change.

Suggested Solutions

Marks

Marker's Comments

(iv) $A = \$344555.87$

$M = \$6661.23$ (monthly repayment).

Let P_n be the amount owing after n repayments.

New interest rate $r = \frac{5\%}{12}$ per month
 $= \frac{0.05}{12}$
 $= \frac{1}{240}$

$P_1 = A \times \left(1 + \frac{1}{240}\right)^1 - M$

$P_2 = \left(A \times \left(\frac{241}{240}\right)^1 - M\right) \left(\frac{241}{240}\right) - M$

$= A \left(\frac{241}{240}\right)^2 - M \left(\frac{241}{240}\right) - M$

$P_3 = A \left(\frac{241}{240}\right)^3 - M \left(\frac{241}{240}\right)^2 - M \left(\frac{241}{240}\right) - M$

⋮

$P_n = A \left(\frac{241}{240}\right)^n - M \left(1 + \frac{241}{240} + \left(\frac{241}{240}\right)^2 + \dots + \left(\frac{241}{240}\right)^{n-1}\right)$

this is a GP where

$a = 1$

$r = \frac{241}{240}$

①*

$$P_n = A \left(\frac{241}{240} \right)^n - M \left(\frac{1 \left(\frac{241}{240} \right)^n - 1}{\frac{1}{240}} \right)$$

$$= A \left(\frac{241}{240} \right)^n - M \left(240 \left[\left(\frac{241}{240} \right)^n - 1 \right] \right)$$

$$= A \left(\frac{241}{240} \right)^n - 240 M \left(\left(\frac{241}{240} \right)^n - 1 \right)$$

The loan is paid off when $P_n = 0$

$$\text{ie } A \left(\frac{241}{240} \right)^n - 240 M \left(\left(\frac{241}{240} \right)^n - 1 \right) = 0 \quad (1)^*$$

$$A \left(\frac{241}{240} \right)^n = 240 M \left(\left(\frac{241}{240} \right)^n - 1 \right)$$

$$A \left(\frac{241}{240} \right)^n = 240 M \left(\frac{241}{240} \right)^n - 240 M$$

$$A \left(\frac{241}{240} \right)^n - 240 M \left(\frac{241}{240} \right)^n = -240 M$$

$$\left(\frac{241}{240} \right)^n (A - 240 M) = -240 M$$

$$\left(\frac{241}{240} \right)^n = \frac{-240 M}{A - 240 M}$$

$$n \ln \left(\frac{241}{240} \right) = \ln \left(\frac{-240 M}{A - 240 M} \right)$$

$$\ln \left(\frac{241}{240} \right)$$

Suggested Solutions

Marks

Marker's Comments

$$\text{Substitute } A = 344555.87 \text{ \& } M = \$6661.23 \\ n = 58.37$$

∴ by the 59th payment the loan has been paid off.

This is 4 yrs & 11 months after January 2022
ie November 2026.

(1)*

(V) There are 118 payments (9 yrs & 10 mths)
of \$6661.23 + the last payment in December 2026.

$$\therefore \text{Repayments} = 118 \times \$6661.23 \\ = \$786025.14.$$

$$\text{After } P_{58} = A \left(\frac{241}{240} \right)^{58} - 240M \left(\left(\frac{241}{240} \right)^{58} - 1 \right) \\ \text{where } A = \$344555 \text{ \& } M = \$6661.23$$

$$P_{58} = \$2513.83$$

This means after the 2nd last payment there is only \$2513.83 left to pay.

∴ the last payment needs to be

$$\$2513.83 \times \left(\frac{241}{240} \right)$$

$$= \$2524.30$$

$$\therefore \text{Total amount paid} \\ = 118 \times 6661.23 + 2524.30$$

$$= \$788549.44$$

$$\therefore \text{Total interest} = \$788549.44 - 600000 \\ = \$188549.44$$

(1)*

(1)*